# MCP + RAG for 1C — one-page summary

Local-first search over BSL business logic and configuration metadata · Sergey Matyukhin

## PROBLEM

- A 1C configuration dump is thousands of .bsl modules plus a multi-megabyte metadata report; grep finds text, not meaning.
- An exact handler name and an intent question need different search strategies, and an LLM agent needs a tool with measured quality.
- Hosted search stacks require deployment work and send the code off the machine.

## APPROACH

- Local MCP server over stdio: 9 tools, including an explainable router and a diagnostic index_stats.
- intfloat/multilingual-e5-small (384-d, 512-token limit) + Chroma with HNSW cosine, CPU only.
- Custom global BM25 inverted index; Reciprocal Rank Fusion with k = 60; optional flashrank cross-encoder, off by default.
- No external APIs and no data leaving the machine; 40-60 ms per query.

## CORPUS

- 154,560 BSL code chunks (one procedure or function per chunk, ADR 0007) + 25,026 metadata chunks from the configuration XML = 179,586 chunks.
- Chunking gates: 99.6% code coverage; 0.3% of chunks over 512 tokens; token p50/p99/max = 179 / 472 / 554.
- Metadata source change (text report → XML): BM25 Hit@3 93.0% → 99.7% (ADR 0008).

## QUALITY — GOLD-SET V2: 70 NATURAL-LANGUAGE QUERIES, FILE-LEVEL RELEVANCE, 4 RUNS, MEAN ± STD

| Configuration | nDCG@10 | Recall@10 | Hit@1 | MRR@10 |
|---|---|---|---|---|
| vector | 0.331 ± 0.028 | 0.511 ± 0.044 | 0.157 ± 0.018 | 0.266 ± 0.026 |
| pearson | 0.383 ± 0.040 | 0.562 ± 0.053 | 0.207 ± 0.038 | 0.313 ± 0.039 |
| hybrid (RRF) | 0.442 ± 0.012 | 0.652 ± 0.024 | 0.221 ± 0.008 | 0.368 ± 0.010 |
| lexical (BM25) | 0.426 | 0.571 | 0.271 | 0.355 |
| hybrid + rerank | 0.117 | 0.157 | 0.086 | 0.102 |

Paired test (v2, 4 runs): hybrid 0.442 vs vector 0.331; Δ = −0.112, 95% CI [−0.184; −0.038], p = 0.0037 → hybrid significantly better; default hybrid confirmed.
Independent v3 hold-out (71 code queries, 3 runs): hybrid 0.217 vs vector 0.208; Δ = −0.010, p = 0.635 → not worse. ADR 0009 (leader shift) rejected; fix in ADR 0010.
Router v3: 58/100; main gap lexical→hybrid. Anchors: 10/10 top-3.

## ENGINEERING

- 74 unit tests; ruff clean, mypy over 22 modules, cyclomatic-complexity gate 143 blocks / avg 4.37 (A) in CI; Python 3.11/3.12; CI green.
- ADR log 0001-0009 and CHANGELOG; router 68/68 on the dev set, 58/100 on the independent v3 hold-out (quoted-name lexical cases fall back to hybrid).
- Limitations: HNSW rebuilt on start (~60-70 s cold start, workaround for chroma#7090); single writer; no incremental indexing; gold set partly synthetic.

## LINKS AND CONTACT

**github.com/Smaturus/1c-mcp-rag-search · smaturus@gmail.com · MCP Python SDK · Chroma · multilingual-e5-small · BM25 · RRF · flashrank**

All numbers come from the repository reports: gold-set v2 (4 runs) and independent v3 hold-out (3 runs).